



Acute effects of resistance training at different range of motions on plantar flexion mechanical properties and force

Yuta Murakami, Andreas Konrad, Kazuki Kasahara, Riku Yoshida, Konstantin Warneke, David G Behm & Masatoshi Nakamura

To cite this article: Yuta Murakami, Andreas Konrad, Kazuki Kasahara, Riku Yoshida, Konstantin Warneke, David G Behm & Masatoshi Nakamura (2024) Acute effects of resistance training at different range of motions on plantar flexion mechanical properties and force, Journal of Sports Sciences, 42:1, 38-45, DOI: [10.1080/02640414.2024.2320522](https://doi.org/10.1080/02640414.2024.2320522)

To link to this article: <https://doi.org/10.1080/02640414.2024.2320522>



Published online: 23 Feb 2024.



Submit your article to this journal [↗](#)



Article views: 152



View related articles [↗](#)



View Crossmark data [↗](#)



Acute effects of resistance training at different range of motions on plantar flexion mechanical properties and force

Yuta Murakami^a, Andreas Konrad^b, Kazuki Kasahara^a, Riku Yoshida^c, Konstantin Warneke^d, David G Behm^e and Masatoshi Nakamura^f 

^aInstitute for Human Movement and Medical Sciences, Niigata University of Health and Welfare, Niigata, Japan; ^bInstitute of Human Movement Science, Sport and Health, University of Graz, Graz, Austria; ^cDepartment of Rehabilitation, Maniwa orthopedic clinic, Niigata, Japan; ^dDepartment for Exercise, Sport and Health, Leuphana University, Lüneburg, Germany; ^eSchool of Human Kinetics and Recreation, Memorial University of Newfoundland, Newfoundland and Labrador, Canada; ^fFaculty of Rehabilitation Sciences, Nishi Kyushu University, Kanzaki, Saga, Japan

ABSTRACT

The effects obtained from resistance training depend on the exercise range of motion (ROM) performed. We aimed to examine the acute effects of different exercise ROM resistance training on the plantar flexor muscles. Eighteen healthy untrained male adults participated in three conditions: calf raises in 1) partial condition [final (short muscle length) partial ROM], 2) full condition (full ROM), and 3) control condition. The ankle dorsiflexion (DF) ROM, passive torque at DF ROM, passive stiffness of muscle-tendon unit, and maximal voluntary isometric contraction (MVC-ISO) torque were measured before and immediately after the interventions. There were significant increases in DF ROM, passive torque at DF ROM, and a decrease in MVC-ISO, but no significant interaction in passive stiffness. Post hoc test, DF ROM demonstrated moderate magnitude increases in the full condition compared to the partial ($p = 0.023$, $d = 0.74$) and control ($p = 0.003$, $d = 0.71$) conditions. Passive torque at DF ROM also showed moderate magnitude increases in the full condition compared to the control condition ($p = 0.016$, $d = 0.69$). MVC-ISO had a moderate magnitude decrease in the full condition compared to the control condition ($p = 0.018$, $d = -0.53$). Resistance training in the full ROM acutely increases joint ROM to a greater extent than final partial ROM, most likely due to stretch tolerance.

ARTICLE HISTORY

Received 20 October 2023
Accepted 12 February 2024

KEYWORDS

Flexibility; passive torque; stretch tolerance; passive stiffness; maximal voluntary isometric contraction

Introduction

Recently, a number of studies suggested that resistance training increases joint range of motion (ROM). Alizadeh et al.'s systematic review and meta-analysis reported that long-term resistance training increases joint ROM (Alizadeh et al., 2023). In addition, Warneke et al. investigated the acute effects of resistance training, and they showed an acute increase in joint ROM after full ROM resistance training for the plantar flexor muscles (Warneke et al., 2022). Therefore, although it has been suggested that full ROM resistance training acutely increases joint ROM, the effects obtained may differ depending on whether the resistance training exercise ROM is full or partial [initial: long muscle lengths, final: long muscle lengths, based on concentric contraction (CON)]. Kawama et al. examined the acute effects of deadlifts performed at different exercise ROMs (i.e., initial or final) on passive property of hamstrings. They reported that a full ROM deadlift decreased the semimembranosus muscle stiffness measured by shear wave elastography (Kawama et al., 2022). However, knee extension ROM increased only in the short muscle length exercise ROM (i.e., final partial ROM) deadlift condition but not in the full ROM, making it difficult to explain this result. As a continuation of this previous study, Kawama et al. reported that the semimembranosus muscle stiffness measured by shear wave elastography decreased (Kawama et al., 2023). Additionally, the knee

extension ROM increased only when the long muscle length exercise ROM (i.e., initial partial ROM), plus the duration was long during the deadlift (Kawama et al., 2023). These findings suggest that resistance training at long muscle lengths, such as full ROM and initial partial ROM, acutely reduces muscle stiffness. However, there is no consistent view on the acute effect of exercise ROM on an increase in joint ROM. Moreover, the only previous studies that used joint ROM or muscle stiffness as an outcome of resistance training at different exercise ROMs focused on hamstrings during deadlift exercises (Kawama et al., 2022, 2023). Therefore, there is no knowledge of the effect of resistance training at different exercise ROMs on joint ROM or passive stiffness for other muscles, such as plantar flexor muscles.

Less ankle dorsiflexion (DF) ROM has been associated with greater medial knee displacement during various tasks (Bell et al., 2008; Sigward et al., 2008). In addition, previous studies have suggested that less DF ROM could be a potential aetiology associated with Achilles tendinopathy (Carcia et al., 2010; Kaufman et al., 1999; Kvist, 1994). Moreover, the calf raise exercise is a common technique to improve the strength and power of plantar flexor muscles (Fujisawa et al., 2015). Hence, investigating the effects of calf raise exercises at different exercise ROMs on DF ROM may provide important insights from the injury prevention perspective. Furthermore, the reason why

previous studies have not reached a unified view on the acute effects of resistance training with different ROM on ROM and passive stiffness may be related to the difficulty of the deadlift exercise. Deadlift exercise requires control of the hip joint and trunk muscles, whereas the calf raise exercise involves only the ankle joint. Therefore, it may be possible to clarify the influence of exercise ROM throughout clarifying the acute changes in joint ROM and passive stiffness immediately after the calf raise at different exercise ROM. In addition, mechanical properties such as muscle stiffness and sensations such as pain and discomfort tolerance (i.e., stretch tolerance) have been reported to be involved as factors in the increase in joint ROM (Weppeler & Magnusson, 2010). Therefore, when an increase in joint ROM is observed, the mechanisms involved in the increase in joint ROM should be investigated.

Thus, we have investigated the acute changes in joint ROM and passive stiffness immediately after the calf raise at different exercise ROM [1-s CON followed by a 3-s eccentric contraction (ECC) (final partial vs. full ROM)] on plantar flexor muscles. We hypothesized that full ROM resistance training would immediately increase joint ROM and decrease muscle stiffness. Based on previous studies, we hypothesized that stretch tolerance is related to increased joint ROM (Kawama et al., 2022).

Methods

Participants

A total of 18 healthy untrained male adults participated in this study (age: 22.9 ± 1.0 years, height: 171.5 ± 4.5 cm, and body weight: 69.2 ± 7.7 kg). The inclusion criterion was as follows: no regular resistance training within the past 6 months. The exclusion criteria were as follows: neuromuscular disease and a history of lower-limb orthopaedic injuries. All participants provided written informed consent after being fully informed of the study procedures and purposes. After performing the sample size calculation (primary outcome: ROM) for a two-way repeated-measures ANOVA [effect size = 0.25 (medium), α error = 0.05, and power = 0.80] using G* power 3.1 software (Heinrich Heine University, Düsseldorf, Germany), based on a previous study (Kawama et al., 2022), it was determined that more than 18 participants were required in this study. This study was conducted following the Declaration of Helsinki

and was approved by the ethics community (ethics approval number: #17677).

Experimental set-up

The study used a crossover and counterbalanced design, and all participants performed all three conditions. 1) the final partial ROM (partial condition), 2) the full ROM (full condition), and 3) rest (control condition) on ankle DF ROM, passive torque at DF ROM, passive stiffness of muscle-tendon unit, and muscle strength in the dominant leg (preferred leg to kick a ball) (Nakamura, Yahata, et al., 2021). Each participant was instructed to visit the laboratory four times with an interval of >72 h. The first day was a familiarization session, including a demonstration of the calf raise intervention and the measurement protocols. It has been demonstrated that the magnitude of muscle damage induced by a session of ECC is higher on the first occasion than on the subsequent occasions, which is referred to as the “repeated bout effect” (Nosaka et al., 2001). To minimize the effects of muscle damage, a familiarization session was conducted on the first day. If the subject complained of pain due to muscle damage on the day of measurement, the recovery interval was prolonged until the pain disappeared.

All conditions were performed in random order in the remaining three sessions. The calf raise intervention (i.e., final partial and full ROM) consisted of a 1-s CON followed by a 3-s ECC. Three sets of 10 repetitions were performed, with a 60-s rest between sets. In the control condition, the participant was instructed to sit for 240 s (Figure 1). All the outcome variables were measured before (PRE) and immediately after the intervention (POST). In both the PRE and POST measurements, 1) DF ROM, passive torque at DF ROM, and passive stiffness of muscle-tendon unit, and 2) maximal voluntary isometric contraction (MVC-ISO) were measured in this order.

Assessment of the DF ROM, passive torque, and passive stiffness

The participant was secured in a seated position on the chair of an isokinetic dynamometer, with a knee angle of 0° (i.e., anatomical position). Moreover, the trunk and pelvis of the participant were fixed with a belt while the participant was reclined

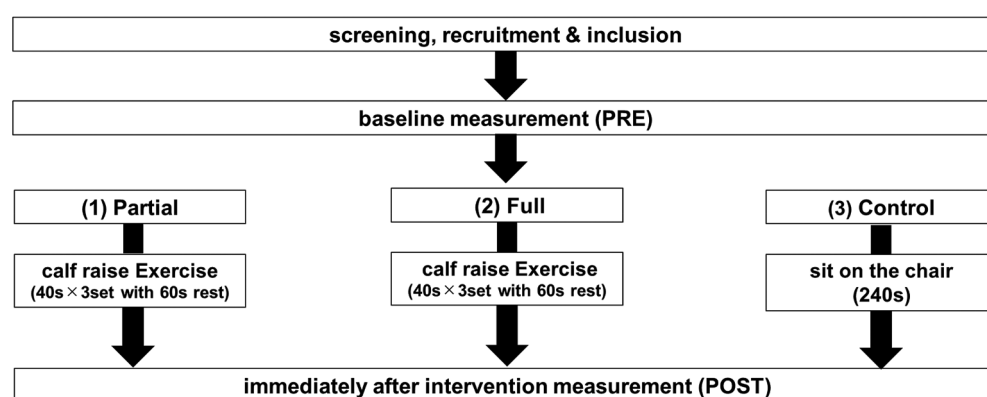


Figure 1. Experimental flowchart.

(with the hip angle at 70°) to prevent tension at the back of the knee (Nakamura et al., 2020). Thereafter, the footplate of the dynamometer was passively and isokinetically dorsiflexed at a speed of 5°/s from the 30° plantar flexion to the dorsiflexion angle just before participants started to feel discomfort or pain (Akagi & Takahashi, 2013). After two familiarization trials, the participant stopped the dynamometer by activating a safety trigger when they started to feel discomfort or pain. The angle was defined as the DF ROM (°) (Nakamura, Yahata, et al., 2021; Nakamura, Yoshida, et al., 2021). The measurement was performed twice, and the average value was used for analysis. In addition, passive plantar flexion torque was measured during the DF ROM measurement, and passive torque at DF ROM (Nm) was defined as the passive plantar flexion torque at the time maximum DF ROM was measured (Fukaya et al., 2020). Passive torque at DF ROM was an indicator of the stretch tolerance (Mizuno et al., 2013; Weppeler & Magnusson, 2010). Furthermore, to confirm that there was no voluntary contraction of the plantar flexor muscles, surface electrodes (Blue Sensor N, Ambu A/S, Ballerup, Denmark) were applied to the medial gastrocnemius muscle, and electromyography (FA-DL-720-140; 4Assist, Tokyo, Japan) was used to confirm muscle activity (Nakamura et al., 2013).

The passive stiffness (Nm/°) was calculated from the passive torque at DF ROM at a certain angle, based on a previous study (Fukaya et al., 2020). For the calculation, the slope of torque to angle from the last 50% of the smallest DF ROM to the maximum angle (100%) among the DF ROMs was measured four times, both PRE and POST. The measurements were repeated twice, and the average of these measurements was used in the analyses.

Assessment of MVC-ISO torque

MVC-ISO torque measurements were made using the dynamometer, with a similar position to the DF ROM assessment. Moreover, the MVC-ISO of the plantar flexors was measured at the ankle joint in the anatomically neutral position (0°) (Nakamura, Sato, et al., 2021). MVC-ISOs were performed for 3 s over two sets, with a 60-s rest between each set. Throughout the measurement, the participant was verbally encouraged by the investigator during the muscle contraction to promote maximum effort. The average MVC-ISO value was used for the analyses.

Calf raise exercise

The calf raise implementation method is shown in Figure 2. The calf raise exercise was performed with one leg supported on the dominant leg. The knee joint of the dominant leg was in extension. The exercise ROM was set from full plantar flexion to neutral position in the partial condition and from full plantar flexion to full DF in the full condition. The calf raise intervention consisted of a 1-s CON followed by a 3-s ECC. Three sets of 10 repetitions with body weight were performed, with a 60-s rest between sets.

Measurement reliability

Eight new participants (mean age: 22.3 ± 1.0 years, height: 170.4 ± 5.8 cm, and body weight: 70.1 ± 10.1 kg) were selected, and test-retest reliability was calculated using test 1 and test 2 (interval of >72 h).

Statistical analysis

Statistical analysis was performed using SPSS (version 28.0; SPSS Japan Inc., Tokyo, Japan). The Shapiro–Wilk test was used to test the normality of the data. The PRE and POST values for each variable followed normality. To analyse the interaction and main effects, all the outcome variables were compared using a 2 × 3 two-way repeated-measures ANOVA (time [PRE vs. POST] and conditions [partial vs. full vs. control]). On confirming a significant interaction effect, changes from PRE to POST values were calculated, and Scheffé tests were conducted as post hoc tests to compare these changes between conditions. Effect sizes are presented as partial eta-squared values (η_p^2) and categorized as either small effect ($\eta_p^2 < 0.01$), medium effect ($\eta_p^2 = 0.02–0.1$), or large effect ($\eta_p^2 > 0.1$) (Cohen, 1988). In addition, effect sizes are reported with Cohen's *d* and categorized as either small effect ($d < 0.5$), medium effect ($d = 0.5–0.8$), or high effect ($d > 0.8$) (Cohen, 1988). The relationship between the change in DF ROM and the change in passive torque at DF ROM, and also the passive stiffness in the partial and full conditions, was examined using the Pearson product-moment correlation coefficient. A *p* value of < 0.05 indicates statistical significance.

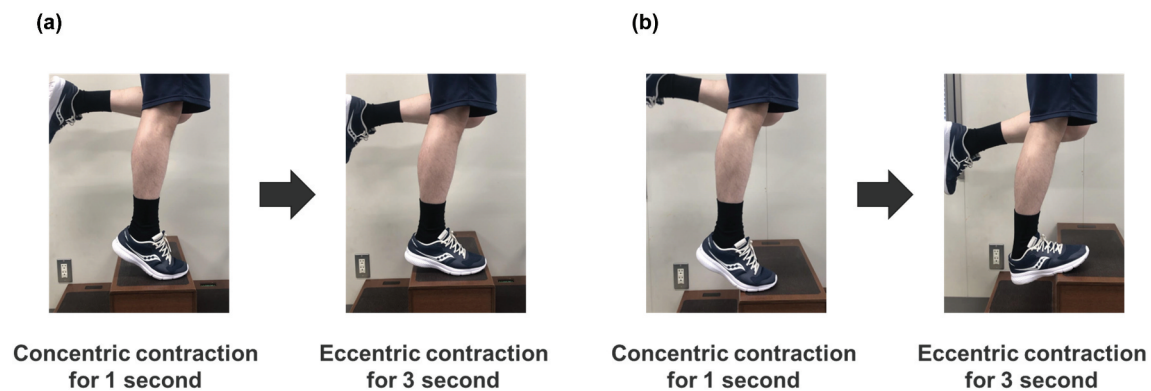


Figure 2. Calf raise intervention set-up for partial (a) and full (b) conditions.

Results

Measurement reliability

The results of the reliability assessments are summarized in Table 1, where it can be seen that high-reliability coefficients were confirmed in this study.

Changes in DF ROM, passive torque at DF ROM, passive stiffness, and MVC-ISO torque

Table 2 lists the results for the DF ROM, passive torque at DF ROM, passive stiffness, and MVC-ISO before and after the intervention in each condition. DF ROM shows a significant main effect for time ($p < 0.001$, $F = 16.1$, $\eta_p^2 = 0.486$) and also an interaction effect ($p = 0.004$, $F = 6.62$, $\eta_p^2 = 0.28$). The post hoc test shows a significant and moderate magnitude greater ROM with the full vs. partial ($p = 0.023$, $d = 0.74$) and control conditions ($p = 0.003$, $d = 0.71$). However, there are no significant

differences between the partial and control conditions ($p = 0.752$, $d = 0.21$) Figure 3(a).

Passive torque at DF ROM shows a significant main effect for time ($p = 0.014$, $F = 8.72$, $\eta_p^2 = 0.34$) and also an interaction effect ($p = 0.009$, $F = 4.87$, $\eta_p^2 = 0.223$). The post hoc test shows significant and moderate magnitude greater passive torque with the full vs. control conditions ($p = 0.016$, $d = 0.69$). However, no significant differences are apparent between the partial and full ($p = 0.167$, $d = 0.54$) and partial and control conditions ($p = 0.574$, $d = 0.32$) Figure 3(a).

Passive stiffness shows no significant main effect for time ($p = 0.839$, $F = 0.04$, $\eta_p^2 = 0.002$) and no interaction effect ($p = 0.251$, $F = 1.44$, $\eta_p^2 = 0.078$).

MVC-ISO torque demonstrates a significant main effect for time ($p < 0.001$, $F = 38.22$, $\eta_p^2 = 0.69$) and also an interaction effect ($p = 0.014$, $F = 4.86$, $\eta_p^2 = 0.222$). The post hoc test reveals significant moderate magnitude greater decreases with the full vs. control conditions ($p = 0.018$, $d = -0.53$). However, there are no significant differences between the partial and full conditions ($p = 0.376$, $d = -0.28$) or the partial and control conditions ($p = 0.781$, $d = -0.20$) Figure 3(c).

The relationship between change in DF ROM and passive torque at DF ROM, and also passive stiffness

Significant positive correlations are found between DF ROM change and passive torque at DF ROM in the partial ($p < 0.001$, $r = 0.818$) and full ($p < 0.001$, $r = 0.876$) conditions. No significant correlation is found between DF ROM change

Table 1. Reliability assessment of dorsiflexion range of motion (DF ROM), passive torque at DF ROM, passive stiffness, and maximal voluntary isometric contraction (MVC-ISO) measurements (mean $\pm$ SD).

	Test1	Test2	ICC(1.1)	95% CI
DF ROM (°)	15.8 $\pm$ 6.2	15.0 $\pm$ 6.7	0.868	0.515–0.972
Passive torque at DF ROM (Nm)	26.5 $\pm$ 7.4	25.1 $\pm$ 6.0	0.887	0.572–0.976
Passive stiffness (Nm/°)	0.69 $\pm$ 0.21	0.64 $\pm$ 0.18	0.870	0.522–0.972
MVC-ISO (Nm)	161.8 $\pm$ 22.3	172.9 $\pm$ 17.4	0.866	0.510–0.971

Table 2. Values and percentage change (mean $\pm$ SD) in the dorsiflexion range of motion (DF ROM), passive torque at DF ROM, passive stiffness, and maximal voluntary isometric contraction (MVC-ISO) before (PRE) and after (POST) the intervention. The two-way repeated-measure ANOVA results (time effect, time $\times$ condition interaction effect; F value) and η_p^2 are shown in Table 2.

	Condition	PRE	POST	Change (%)	Time effect	Time $\times$ condition
DF ROM (°)	Partial	17.9 $\pm$ 9.2	19.1 $\pm$ 9.6	2.3 $\pm$ 3.6	$p < 0.001$	$p = 0.004$
	Full	16.9 $\pm$ 8.8	20.5 $\pm$ 10.3	7.6 $\pm$ 6.1	$F = 16.1$	$F = 6.62$
	Control	17.4 $\pm$ 8.5	17.8 $\pm$ 9.1	0.9 $\pm$ 6.0	$\eta_p^2 = 0.486$	$\eta_p^2 = 0.28$
Passive torque at DF ROM (Nm)	Partial	27.4 $\pm$ 10.3	29.4 $\pm$ 10.3	8.4 $\pm$ 15.3	$p = 0.009$	$p = 0.014$
	Full	26.2 $\pm$ 9.3	31.3 $\pm$ 11.7	20.2 $\pm$ 19.0	$F = 8.72$	$F = 4.87$
	Control	27.4 $\pm$ 9.5	28.3 $\pm$ 11.3	1.9 $\pm$ 19.1	$\eta_p^2 = 0.002$	$\eta_p^2 = 0.223$
Passive stiffness (Nm/°)	Partial	0.67 $\pm$ 0.24	0.66 $\pm$ 0.21	−0.5 $\pm$ 6.7	$p = 0.839$	$p = 0.251$
	Full	0.65 $\pm$ 0.24	0.64 $\pm$ 0.22	0.4 $\pm$ 9.4	$F = 0.04$	$F = 1.44$
	Control	0.67 $\pm$ 0.26	0.68 $\pm$ 0.26	2.6 $\pm$ 6.4	$\eta_p^2 = 0.339$	$\eta_p^2 = 0.078$
MVC-ISO (Nm)	Partial	162.6 $\pm$ 27.4	155.6 $\pm$ 24.7	−4.0 $\pm$ 4.5	$p < 0.001$	$p = 0.014$
	Full	170.3 $\pm$ 24.3	159.5 $\pm$ 23.8	−6.3 $\pm$ 5.6	$F = 38.22$	$F = 4.86$
	Control	168.1 $\pm$ 27.9	166.0 $\pm$ 29.3	−2.1 $\pm$ 6.9	$\eta_p^2 = 0.692$	$\eta_p^2 = 0.222$

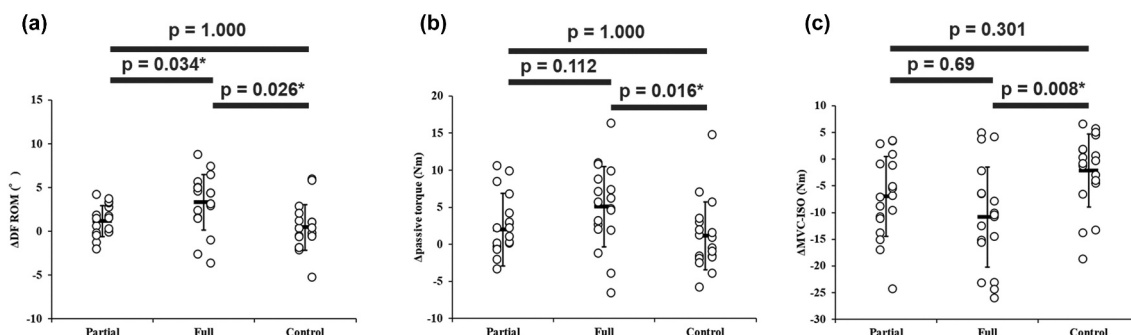


Figure 3. Changes (%) in dorsiflexion range of motion (DF ROM) (a), passive torque at DF ROM (b), and maximal voluntary isometric contraction (MVC-ISO) (c) in partial, full, and control conditions. * $p < 0.05$.

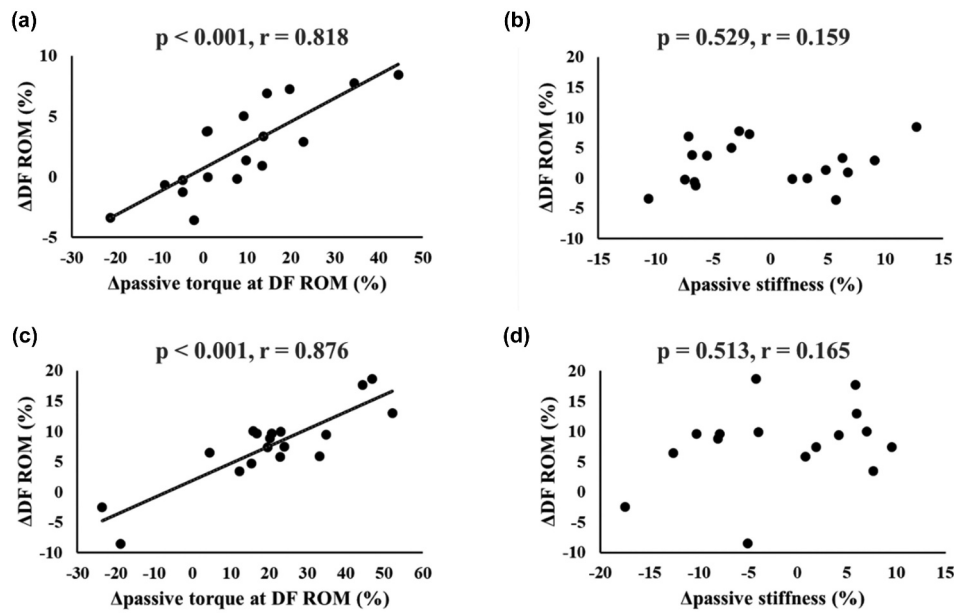


Figure 4. Relationship between change (%) in dorsiflexion range of motion (DF ROM) and change in passive torque at DF ROM (a) and passive stiffness (b) in the partial condition. Relationship between change in DF ROM and change in passive torque at DF ROM (c) and passive stiffness (d) in the full condition.

and passive stiffness change in the partial ($p = 0.529$, $r = 0.159$) and full ($p = 0.513$, $r = 0.165$) conditions (Figure 4).

Discussion

In this study, we examined the acute effects of the calf raise exercise consisting of a 1-s CON followed by a 3-s ECC at different ROMs on passive properties and muscle strength in the plantar flexor muscles. In accordance with the hypothesis, DF ROM was significantly increased in the full condition compared to the partial and control conditions. Passive torque at DF ROM was significantly increased, whereas MVC-ISO was significantly decreased in the full compared to the control condition. In contrast to the hypothesis, there were no significant differences in passive stiffness between conditions.

DF ROM and passive torque at DF ROM experienced moderate magnitude, significant increases in the full condition, with no significant differences between the partial and control conditions. In addition, significant positive correlations were found between DF ROM change and passive torque at DF ROM in the partial and full conditions (Figure 4). However, no significant correlation was found between changes in DF ROM and passive stiffness in the partial and full conditions. Since passive torque at the time of maximum ROM measurement is an indicator of stretch tolerance (Mizuno et al., 2013; Weppeler & Magnusson, 2010), the increase in joint ROM due to the acute effect of partial and full conditions might be due to an increase in stretch tolerance rather than changes in the passive mechanical properties of the muscle in this study. In the previous study examining the acute effects of resistance training (Kawama et al., 2022, 2023), changes in knee extension ROM showed a significant positive correlation with changes in passive torque in the final position but no significant correlation with changes in semimembranosus, biceps femoris long head, and semitendinosus stiffness measured by shear wave elastography. The

results of this study are consistent with previous studies (Kawama et al., 2022, 2023) examining acute effects. In this study, the partial condition did not significantly increase passive torque in DF ROM compared to the control condition. However, a significant positive correlation between changes in DF ROM and changes in passive torque at DF ROM was also observed in the partial condition. Therefore, the mechanism for joint ROM increase in both the partial and full conditions could be stretch tolerance. The fact that the full condition showed a significantly greater increase in joint ROM than the partial condition could be related to the intensity of the mechanical stress applied to the muscle-tendon unit. A previous study of 180s of static stretching of the hamstrings at different intensities (80%, 100%, and 120%, with 100% being the maximum tolerable range without stretching pain) reported significant increases in joint ROM only in the 100% and 120% intensity static stretching conditions (Kataura et al., 2017). Therefore, the higher the mechanical stress applied to the muscle-tendon unit, the greater the effect of increased stretch tolerance should be. Note, however, that the partial condition in this study was performed with a short muscle length. The effect of a short muscle length and a long muscle length in the same partial may be different. In the future, it is necessary to examine whether resistance training at any exercise ROM, including partial ROM resistance training with longer muscle length, is effective for joint ROM and other flexibility.

The results of this study showed that passive stiffness did not change significantly in all conditions. In the previous study by Kawama et al. comparing the acute effects of resistance training at different exercise ROMs on hamstrings, they reported that the semitendinosus stiffness measured by shear wave elastography was significantly reduced only under conditions at full ROM (Kawama et al., 2022). Therefore, the results suggest that muscle stiffness is rapidly decreased by resistance training at full ROM. However, in the present study, passive

stiffness was not significantly changed in all conditions, which is inconsistent with the previous study. One possible factor is that the muscles (joints) of interest differed between the previous and present studies. The ankle joint has a smaller absolute ROM than the knee joint, and the warming-up effect is less effective, so passive stiffness may not have decreased in this study. In fact, a previous study reported that dynamic stretching decreased passive stiffness in the hamstrings (Iwata et al., 2019), whereas it did not decrease passive stiffness in the plantar flexors (Mizuno & Umemura, 2016). These results support the idea that the absolute ROM differs between the ankle and knee joints; therefore, the warm-up effect is different. In addition, the resistance training in the previous study by Kawama et al. consisted of ECC-only (Kawama et al., 2022), while the resistance training in the present study consisted of both CON and ECC. Therefore, ECC is important to reduce muscle stiffness, and it may be necessary to perform resistance training with ECC-only.

MVC-ISO showed a moderate magnitude significant decrease in the full condition compared to the control condition. However, there was no significant difference between changes in partial and control conditions. Muscle fatigue contributes to muscle decrease. Sustained physical exertion leads to a reduced capacity to produce voluntary force (fatigue), and fatigue is caused by both a reduction in muscle function, called peripheral fatigue, and a reduction in the central nervous system's ability to activate muscles, called central fatigue (Carroll et al., 2017). Resistance training in the full ROM may induce greater muscle fatigue. Mendonça et al. compared force generation in male athletes of elite national level in Paralympic Powerlifting by bench pressing at full ROM and final partial ROM (short muscle length) (Mendonça et al., 2021). The results showed a greater decrease in maximum isometric force in the full ROM ($d=3.53$) than in the final partial ROM ($d=1.85$), suggesting that resistance training in the full ROM induces greater signs of muscle fatigue. Therefore, resistance training in the full ROM may induce greater muscle fatigue and an acute decrease in muscle strength than resistance training at the partial ROM. However, there were no significant differences between the partial and full conditions in this study. There may not have been much difference in calf raise ROM between the partial and full conditions due to the small absolute ROM of the ankle joint. In the present study, there was no significant difference between the partial and full conditions in terms of muscle fatigue, and it is possible that there was no difference in the degree of decrease in MVC-ISO.

Previous studies pointed out that the exercise ROM of resistance training is one of the variables that also influences the increase in muscle strength and muscle mass with long-term training. Pallarés et al. reported in their review that routine full ROM resistance training resulted in greater gains in muscle strength and lower-limb muscle hypertrophy compared to partial ROM resistance training (Pallarés et al., 2021). Also, a systematic review by Kassiano et al. suggests that performing routine resistance training with long muscle lengths throughout the partial or full ROM is an important factor in muscle hypertrophy (Kassiano, Costa, Nunes, et al., 2023). Furthermore, Kassiano et al. performed 8 weeks of resistance training on the gastrocnemius muscle at different ROMs: full ROM (ankle joint:

-25° to $+25^{\circ}$), initial ROM [maximum ankle joint dorsiflexion: -25° to 0° based on CON], and final ROM (maximum ankle joint plantar flexion: 0° to $+25^{\circ}$ based on CON) (Kassiano, Costa, Kunevaliki, et al., 2023). The results showed that muscle thickness of the medial and lateral gastrocnemius muscles increased significantly only in the group that performed resistance training at full ROM and initial ROM. The increase in muscle thickness of the medial gastrocnemius muscle was greater in the initial ROM group than in the full ROM group. Thus, resistance training at long muscle lengths, such as full ROM and initial partial ROM is important for increasing muscle strength and muscle hypertrophy. The results of this study suggest that full ROM resistance training is more effective than final partial ROM resistance training in increasing acute joint ROM. Therefore, resistance training at longer muscle lengths, such as full ROM and initial partial ROM, may be important to increase joint ROM, as well as in chronic effects. In the future, the effect of routine resistance training at different ROM on joint ROM should be investigated.

There are several limitations to this study. An important limitation of this study is that the calf raise exercise was performed with body weight, so the intensity of the exercise may have varied among the participants. In the future, it is necessary to change the exercise method and the method of applying the load (weights) and standardize the load's intensity among the participants. In addition, healthy, untrained male adults without lower-extremity resistance training habits were included in this study. Therefore, it is impossible to say whether similar results could be obtained for experienced resistance training individuals. In addition, this study did not use ultrasonography or reflective markers to measure muscle length or DF angle during the calf raises. Therefore, it is impossible to say to what extent muscle fascicle length and DF angle actually differed between the partial and full conditions. The calf raise exercise employed in this study consisted of a 1-s CON followed by a 3-s ECC on one leg. Kawama et al. reported a significant increase in knee extension ROM with ECC in a narrow exercise ROM with a duration of 2-s (Kawama et al., 2022). In addition, Kawama et al. reported that ECC of long duration (5-s) at long muscle length significantly increased knee extension ROM (Kawama et al., 2023). These findings suggest that ECC contraction duration (i.e., the velocity of the movement) and muscle length could have contributed to the changes in joint ROM. However, we investigated the acute effect of only one duration (1-s CON followed by a 3-s ECC). It consisted of both CON and ECC contraction modes. Therefore, contraction duration (speed) and contraction mode could affect the results of DF ROM in this study. Thus, future studies are needed to investigate the effects of different contraction durations (speeds), contraction modes, and muscle lengths on joint ROM and passive stiffness of gastrocnemius muscles. In this study, both intervention conditions resulted in joint ROM increases, which have also been reported in response to several other training conditions, such as foam rolling (Nakamura, Onuma, et al., 2021), cycling (Morales-Artacho et al., 2017), heat application (Nakamura et al., 2023), and full ROM resistance training (Warneke et al., 2022), so that warm-up effects cannot be excluded as an explanatory

mechanism. Therefore, future studies should be conducted to improve on the limitations identified in this article and rule out warm-up effects by comparing the effects with those of other approaches. Muscle and tissue temperature of the intervened muscles should also be investigated.

Conclusion

This study compared the acute effects on plantar flexors' passive properties and muscle strength between two different exercise ROMs during calf raise exercises and control without exercise. The results suggested that full ROM calf raises effectively increase DF ROM and stretch tolerance compared to final partial ROM (short muscle length) calf raises but do not decrease the passive stiffness of the muscle-tendon unit. The results also suggested that calf raises in the final partial ROM and full ROM immediately decrease muscle strength to a similar degree. Acute full ROM resistance training seems to provide a greater joint ROM increase than final partial ROM resistance training.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by JSPS KAKENHI with grant number [19K19890] (Masatoshi Nakamura) and the Austrian Science Fund (FWF) project [J4484] (Andreas Konrad).

ORCID

Masatoshi Nakamura  <http://orcid.org/0000-0002-8184-1121>

Author contributions

YM, AK, KW, DB, and MN designed the study. YM, KK, and RY were responsible for the data acquisition, and YM and MN did the analysis; the data interpretation was undertaken by YM, AK, KW, DB, and MN; all authors were involved in drafting and critically revising the manuscript for intellectual content. All authors read and approved the final manuscript.

Research involving human participants

All procedures in the study were in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. This study was approved by the Ethics Committee of Niigata University of Health and Welfare (#17677).

References

- Akagi, R., & Takahashi, H. (2013). Acute effect of static stretching on hardness of the gastrocnemius muscle. *Medicine and Science in Sports and Exercise*, 45(7), 1348–1354. <https://doi.org/10.1249/MSS.0b013e3182850e17>
- Alizadeh, S., Daneshjoo, A., Zahiri, A., Anvar, S. H., Goudini, R., Hicks, J. P., Konrad, A., & Behm, D. G. (2023). Resistance training induces improvements in range of motion: A systematic review and meta-analysis. *Sports Medicine*, 53(3), 707–722. © 2023. The Author(s). New Zealand. <https://doi.org/10.1007/s40279-022-01804-x>
- Bell, D. R., Padua, D. A., & Clark, M. A. (2008). Muscle strength and flexibility characteristics of people displaying excessive medial knee displacement. *Archives of Physical Medicine and Rehabilitation*, 89(7), 1323–1328. <https://doi.org/10.1016/j.apmr.2007.11.048>
- Carcia, C. R., Martin, R. L., Houck, J., & Wukich, D. K. (2010). Achilles pain, stiffness, and muscle power deficits: Achilles tendinitis. *The Journal of Orthopaedic and Sports Physical Therapy*, 40(9), A1–26. <https://doi.org/10.2519/jospt.2010.0305>
- Carroll, T. J., Taylor, J. L., & Gandevia, S. C. (2017). Recovery of central and peripheral neuromuscular fatigue after exercise. *Journal of Applied Physiology*, 122(5), 1068–1076. <https://doi.org/10.1152/jappphysiol.00775.2016>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Hillsdale: Routledge.
- Fujisawa, H., Suzuki, H., Nishiyama, T., & Suzuki, M. (2015). Comparison of ankle plantar flexor activity between double-leg heel raise and walking. *Journal of Physical Therapy Science*, 27(5), 1523–1526. <https://doi.org/10.1589/jpts.27.1523>
- Fukaya, T., Nakamura, M., Sato, S., Kiyono, R., Yahata, K., Inaba, K., Nishishita, S., & Onishi, H. (2020). The relationship between stretching intensity and changes in passive properties of gastrocnemius muscle-tendon unit after static stretching. *Sports*, 8(11), 140. <https://doi.org/10.3390/sports8110140>
- Iwata, M., Yamamoto, A., Matsuo, S., Hatano, G., Miyazaki, M., Fukaya, T., Fujiwara, M., Asai, Y., & Suzuki, S. (2019). Dynamic stretching has sustained effects on range of motion and passive stiffness of the hamstring muscles. *Journal of Sports Science and Medicine*, 18(1), 13–20.
- Kassiano, W., Costa, B., Kunevaliki, G., Soares, D., Zacarias, G., Manske, I., Takaki, Y., Ruggiero, M. F., Stavinski, N., Francsuel, J., Tricoli, I., Carneiro, M. A. S., & Cyrino, E. S. (2023). Greater gastrocnemius muscle hypertrophy after partial range of motion training performed at long muscle lengths. *Journal of Strength & Conditioning Research*, 37(9), 1746–1753. <https://doi.org/10.1519/jsc.0000000000004460>
- Kassiano, W., Costa, B., Nunes, J. P., Ribeiro, A. S., Schoenfeld, B. J., & Cyrino, E. S. (2023). Which ROMs lead to Rome? A systematic review of the effects of range of motion on muscle hypertrophy. *Journal of Strength & Conditioning Research*, 37(5), 1135–1144. <https://doi.org/10.1519/jsc.0000000000004415>
- Kataura, S., Suzuki, S., Matsuo, S., Hatano, G., Iwata, M., Yokoi, K., Tsuchida, W., Banno, Y., & Asai, Y. (2017). Acute effects of the different intensity of static stretching on flexibility and isometric muscle force. *Journal of Strength & Conditioning Research*, 31(12), 3403–3410. <https://doi.org/10.1519/JSC.0000000000001752>
- Kaufman, K. R., Brodine, S. K., Shaffer, R. A., Johnson, C. W., & Cullison, T. R. (1999). The effect of foot structure and range of motion on musculoskeletal overuse injuries. *The American Journal of Sports Medicine*, 27(5), 585–593. <https://doi.org/10.1177/03635465990270050701>
- Kawama, R., Hojo, T., & Wakahara, T. (2023). Acute changes in passive stiffness of the individual hamstring muscles induced by resistance exercise: Effects of muscle length and exercise duration. *European Journal of Applied Physiology*, 123(3), 655–666. <https://doi.org/10.1007/s00421-022-05092-1>
- Kawama, R., Yanase, K., Hojo, T., & Wakahara, T. (2022). Acute changes in passive stiffness of the individual hamstring muscles induced by resistance exercise: Effects of contraction mode and range of motion. *European Journal of Applied Physiology*, 122(9), 2071–2083. <https://doi.org/10.1007/s00421-022-04976-6>
- Kvist, M. (1994). Achilles tendon injuries in athletes. *Sports Medicine*, 18(3), 173–201. <https://doi.org/10.2165/00007256-199418030-00004>
- Mendonça, T. P., Aidar, F. J., Matos, D. G., Souza, R. F., Marçal, A. C., Almeida-Neto, P. F., Cabral, B. G., Garrido, N. D., Neiva, H. P., Marinho, D. A., Marques, M. C., Reis, V. M., & Pessôa Filho, D. M. (2021). Force production and muscle activation during partial vs. full range of motion in paralympic powerlifting. *Public Library of Science ONE*, 16(10), e0257810. <https://doi.org/10.1371/journal.pone.0257810>
- Mizuno, T., Matsumoto, M., & Umemura, Y. (2013). Viscoelasticity of the muscle-tendon unit is returned more rapidly than range of motion after stretching. *Scandinavian Journal of Medicine & Science in Sports*, 23(1), 23–30. <https://doi.org/10.1111/j.1600-0838.2011.01329.x>

- Mizuno, T., & Umemura, Y. (2016). Dynamic stretching does not change the stiffness of the muscle-tendon unit. *International Journal of Sports Medicine*, 37(13), e12. <https://doi.org/10.1055/s-0042-118268>
- Morales-Artacho, A. J., Lacourpaille, L., & Guilhem, G. (2017). Effects of warm-up on hamstring muscles stiffness: Cycling vs foam rolling. *Scandinavian Journal of Medicine & Science in Sports*, 27(12), 1959–1969. <https://doi.org/10.1111/sms.12832>
- Nakamura, M., Ikezoe, T., Takeno, Y., & Ichihashi, N. (2013). Time course of changes in passive properties of the gastrocnemius muscle-tendon unit during 5 min of static stretching. *Manual Therapy*, 18(3), 211–215. <https://doi.org/10.1016/j.math.2012.09.010>
- Nakamura, M., Ishikawa, T., Sato, S., Kiyono, R., Yoshida, R., Morishita, K., & Konrad, A. (2023). Time-course changes in Dorsiflexion range of motion, stretch tolerance, and shear elastic modulus for 20 minutes of hot pack application. *Journal of Sports Science and Medicine*, 22(2), 175–179. <https://doi.org/10.52082/jssm.2023.175>
- Nakamura, M., Onuma, R., Kiyono, R., Yasaka, K., Sato, S., Yahata, K., Fukaya, T., & Konrad, A. (2021). The acute and prolonged effects of different durations of foam rolling on range of motion, muscle stiffness, and muscle strength. *Journal of Sports Science and Medicine*, 20(1), 62–68. <https://doi.org/10.52082/jssm.2021.62>
- Nakamura, M., Sato, S., Hiraizumi, K., Kiyono, R., Fukaya, T., & Nishishita, S. (2020). Effects of static stretching programs performed at different volume-equated weekly frequencies on passive properties of muscle-tendon unit. *Journal of Biomechanics*, 103, 109670. <https://doi.org/10.1016/j.jbiomech.2020.109670>
- Nakamura, M., Sato, S., Kiyono, R., Yoshida, R., Murakami, Y., Yasaka, K., Yahata, K., & Konrad, A. (2021). Acute effect of vibration roller with and without rolling on various parts of the plantar flexor muscle. *Frontiers in Physiology*, 12, 716668. <https://doi.org/10.3389/fphys.2021.716668>
- Nakamura, M., Yahata, K., Sato, S., Kiyono, R., Yoshida, R., Fukaya, T., Nunes, J. P., & Konrad, A. (2021). Training and detraining effects following a static stretching program on medial gastrocnemius passive properties. *Frontiers in Physiology*, 12, 656579. <https://doi.org/10.3389/fphys.2021.656579>
- Nakamura, M., Yoshida, R., Sato, S., Yahata, K., Murakami, Y., Kasahara, K., Fukaya, T., Takeuchi, K., Nunes, J. P., & Konrad, A. (2021). Comparison between high- and low-intensity static stretching training program on active and passive properties of plantar flexors. *Frontiers in Physiology*, 12, 796497. <https://doi.org/10.3389/fphys.2021.796497>
- Nosaka, K., Sakamoto, K., Newton, M., & Sacco, P. (2001). The repeated bout effect of reduced-load eccentric exercise on elbow flexor muscle damage. *European Journal of Applied Physiology*, 85(1–2), 34–40. <https://doi.org/10.1007/s004210100430>
- Pallarés, J. G., Hernández-Belmonte, A., Martínez-Cava, A., Vetrovsky, T., Steffl, M., & Courel-Ibáñez, J. (2021). Effects of range of motion on resistance training adaptations: A systematic review and meta-analysis. *Scandinavian Journal of Medicine & Science in Sports*, 31(10), 1866–1881. <https://doi.org/10.1111/sms.14006>
- Sigward, S. M., Ota, S., & Powers, C. M. (2008). Predictors of frontal plane knee excursion during a drop land in young female soccer players. *The Journal of Orthopaedic and Sports Physical Therapy*, 38(11), 661–667. <https://doi.org/10.2519/jospt.2008.2695>
- Warneke, K., Wohllann, T., Lohmann, L. H., Wirth, K., & Schiemann, S. (2022). Acute effects of long-lasting stretching and strength training on maximal strength and flexibility in the calf muscle. *German Journal of Exercise and Sport Research*, 53, 148–154. <https://doi.org/10.1007/s12662-022-00854-7>
- Weppeler, C. H., & Magnusson, S. P. (2010). Increasing muscle extensibility: A matter of increasing length or modifying sensation? *Physical Therapy*, 90(3), 438–449. <https://doi.org/10.2522/ptj.20090012>